

Predicting Ratings from Text: A comparative Regression Experiment

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19 August 2026

1 Introduction

I look at how to automatically predict review ratings using the text from reviews. This approach can support quality monitoring, recommendations, and analytics on consumer platforms. I frame this as a supervised regression problem, aiming to predict a rating from 1 to 5 using the review title and body.

I think that contextual transformer models will predict ratings more accurately than sparse n-gram features, especially for reviews with complex or subtle sentiment. Still, I expect that a traditional TF-IDF approach will remain competitive and efficient.

I compare two methods: a traditional baseline that uses TF-IDF features with Ridge regression, and a neural approach that fine-tunes a transformer model for regression (DistilBERT (Hugging Face, n.d.)). For our experiments, I use the Wine Reviews dataset from Kaggle (krrai77, n.d.).

2 Related Work

Text-based rating prediction has been widely studied in the areas of sentiment analysis and opinion mining. (Pang and Lee, 2005) demonstrated that employing n-gram features with linear classifiers results in good accuracy when classifying sentiment. (Salton and Buckley, 1988) discovered that including review metadata enhances rating prediction. More recently, transformer-based models such as BERT (Devlin et al., 2019) and DistilBERT (Sanh et al., 2019) have obtained excellent results on sentiment tasks by making use of contextual embeddings. In this paper, we carry out a controlled comparison of the various approaches by using a regression approach for the task.

3 Data and Preprocessing

The dataset consists of 2,890 wine reviews with a numerical rating and free-text fields. After removing rows with missing target values, 2,445 samples remain. Table 1 provides a frequency overview.

Table 1: Dataset overview.

Property	Value
Total samples (after cleaning)	2,445
Input features (text)	1 column (Title + Body concatenated)
TF-IDF features (classical)	10,000
Transformer hidden size (neural)	768
Target: mean / std	4.69 / 0.84
Rating distribution: 1 / 2 / 3 / 4 / 5	69 / 43 / 65 / 221 / 2,047
Train / Val / Test split	1,711 / 367 / 367

Table 1: Dataset overview.

The target variable is Reviews Rating. We construct a single text input by concatenating Reviews Title and Reviews Text. Minimal text normalization is applied (lowercasing, whitespace normalization, and HTML tag removal). Rows with missing target values are removed (targets are not imputed).

We use a 70/15/15 train/validation/test split with stratification on the target to preserve rating distribution across splits (seed=42). Model selection is performed on the validation set only, and final metrics are computed once on the held-out test set.

4 Methods

4.1 Baseline: Dummy Regressor

A dummy model predicting the training set mean rating serves as a lower bound.

4.2 TF-IDF + Ridge Regression

Text is represented using TF-IDF weighted unigram and bigram features (max 20,000 features, min document frequency 2). Ridge regression (L2 regularization) is used with hyperparameters tuned via 5-fold cross-validation on the training set.

4.3 DistilBERT Fine-Tuning for Regression

We fine-tune DistilBERT-base-uncased (Sanh et al., 2019) with a single-output regression head. Inputs are tokenized with maximum length 256. Training uses AdamW ($\text{lr} = 2 \times 10^{-5}$, 2 epochs, batch size 8, weight decay 0.01). The best checkpoint is selected by validation MAE.

4.4 Evaluation Metrics

We use Mean Absolute Error (MAE, primary metric, directly interpretable as average rating error in stars), Root Mean Squared Error (RMSE, penalizes large

errors), and R^2 (explained variance).

5 Experiments and Results

Table 2 reports test-set performance

Table 2: Test-set performance on the wine review rating prediction task.

Model	MAE ↓	RMSE ↓	R^2 ↑
Dummy (mean predictor)	0.546	0.916	−0.002
TF-IDF + Ridge	0.461	0.785	0.263
DistilBERT (regression)	0.283	0.657	0.484

The DistilBERT model outperforms Ridge on all three metrics, confirming our hypothesis that contextual representations capture more nuanced sentiment signals than bag-of-words features.

6 Error Analysis and Discussion

The Ridge model’s largest errors all occur for strongly negative reviews (true rating=1.0) predicted as highly positive (4–5). This reveals that linear TF-IDF models struggle with negation (“not very tasty”), sarcasm, and compositional meaning. The transformer model handles these cases more gracefully, though errors persist for short reviews and mixed sentiment.

The transformer requires significantly more computation (5 min training vs. seconds for Ridge), creating a practical trade-off: TF-IDF Ridge offers fast, cheap predictions; DistilBERT provides substantially better accuracy at higher cost.

7 Conclusion

We conducted a methodologically correct comparison between classical and neural regression for rating prediction. The classical TF-IDF Ridge model provides a strong, efficient baseline, while DistilBERT fine-tuning offers superior accuracy. Future work could explore ordinal regression objectives, data augmentation for underrepresented ratings, or larger transformer architectures.

8 Use of AI Tools

Use of AI Tools: ChatGPT was used as a programming assistant for code structure suggestions and lexical correction. All experimental design, execution, and result verification were performed by the author.

References

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